

Cycles 49-51 Report — Sequential-Mode Pickup of M-EAR-1/real-label-training-v2 (Post-c44 Merge Integration)

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Abstract

Cycles 49-51 open the sequential-mode pickup of M-EAR-1/real-label-training-v2 that was PIVOT-escalated to unfixable-by-audit at c42 clone-1 close and awaited root-conductor sequential-mode execution. Cycle 49 was a Hold-Pattern researcher brief (“standing by for c45 fanout directive; no in-flight work items on this clone”) — flagged MINOR by the cycle-51 auditor as an authoring anti-pattern (directive authority is granted by the campaign prompt; no user to wait on). Cycle 50 (worker) proactively advanced on the c26 Path B pre-registered plan and began substantive feature extraction (49 newly-cached clips across 9 songs, in addition to the 203/252 preserved cache from c41-c42). Cycle 51 (worker) is a mid-extraction checkpoint declaring background compute in flight; **auditor decision at cycle 51: CONTINUE** (deliverable absent; not a null cycle; real substantive work running). MODERATE flag on rubric-ordering risk raised proactively for the next cycle: rubric SHA must land BEFORE evaluate_sb_v2 to preserve the 5+ consecutive cycle pre-registration discipline.

Verdict

IN-PROGRESS → **CONTINUE** (extraction advancing; rubric-first pre-registration not yet committed; no verdict emitted; direction sound).

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
49	Hold Pattern brief (“standing by for c45 fanout directive”); no substantive directive	(no worker session invoked)	(no audit)
50	(no researcher brief)	Proactive advance on c26 Path B pre-registered plan; extraction started on 49 newly-cached clips across 9 songs	(no audit)
51	Continue extraction	Mid-extraction checkpoint declaring background compute in flight	CONTINUE (deliverable absent; not a null cycle; MODERATE flag on rubric-ordering risk)

Deliverable State (Cycle 51 Snapshot)

No verdict JSON, no rubric SHA commit event, no ledger events, no test suite, no report exist yet. There is nothing to validate substantively; the worker is blocked on a bash background task completing.

- **CRITICAL findings:** 0
- **MODERATE findings:** 0 (but rubric-ordering risk pre-emptively flagged for next cycle; see below)
- **MINOR findings:** 2 process observations (Hold Pattern brief; researcher-brief absence at cycle 50)

Rubric-Ordering Risk (MODERATE Flag; Correction Required Before evaluate_sb_v2)

Worker’s next-action ordering — “run train_v2 + evaluate_sb_v2, register plan rows, emit ledger events, write report” — **omits rubric-SHA-committed-BEFORE-analysis**, which is the discipline the campaign has held for 5+ consecutive cycles (c26/c27/c28/c29/c30, then c36/c38/c44).

If rubric commit lands after evaluation, this cycle breaks the pre-registration streak. Correction required before evaluate_sb_v2 runs:

1. **Register the milestone row.** Add M-EAR-1/real-label-training-v2 (peer sub-milestone under M-EAR-1; NOT child of terminal-validated v0/v1 per c29 state-machine lemma) to the plan-of-record with full success criteria referencing c26-frozen SB1 margin > 0.5909, SB2 mean $\tau \geq 0.4$, SB3 detection ≥ 0.90 verbatim.
2. **Commit the rubric doc.** Write docs/ear_real_label_training_v2_rubric.md with the 3-verdict frozen rubric (EAR_v2_LANDS / EAR_v2_PARTIAL / EAR_v2_INSUFFICIENT);

embed doc SHA-256 in data/ear_v2/rubric_hash.txt; emit M-EAR-1/real-label-training-v2/rubric-committed ledger event referencing the SHA. **Only THEN run evaluate_sb_v2.** Add a git-mtime + git-log ordering test — this is how c26-c44 held the discipline.

3. **Corpus-N caveat surfaced prominently.** Extraction covers a subset of the 80-song target; report must name exact N and reproduce the c36/c38 “preview / partial corpus” caveat pattern. Do not overstate coverage.
4. **Anchor preservation manifest.** c6 feature cache; c22 stability harness; c26 Path B commitment doc; c36 v0; c38 v1 artefacts remain READ-ONLY. Emit anchor-preservation-verified event with per-file SHA pre/post.
5. **Housekeeping pair.** Standard _archive/cycle-45-scratch + _infra/adopt-cycle45-tests events at cycle tail per codified pattern.

Note on rubric-doc handling: the c42 closure preserved docs/ear_real_label_training_v2_rubric.md and data/ear_v2/rubric_hash.txt untouched (SHA 01948b6efe6ca5e9..., 9,304 B). If those anchors satisfy the new sequential-mode rubric requirements, this cycle should reference them explicitly (documenting continuity) rather than re-authoring. If a new rubric SHA is committed, the c42 SHA snapshot should be logged as a superseded predecessor with rationale.

Sub-Topic Assessment (Zero Criteria Evaluable This Range)

M-EAR-1/real-label-training-v2 (implied — worker never registered a plan row or rubric doc this cycle, so exact milestone id and rubric are unfixed):

Criterion	Status
(a) rubric committed BEFORE analysis	NOT YET (worker’s checkpoint schedules rubric commit + plan-row registration after extraction completes — ordering wrong)
(b) feature extraction complete	IN FLIGHT (49 additional clips advancing on top of preserved 203/252 c41-c42 cache)
(c) train_v2 + evaluate_sb_v2 run	NOT STARTED
(d) SB1/SB2/SB3 thresholds honoured verbatim	NOT YET APPLICABLE
(e) byte-determinism × 2	NOT YET APPLICABLE
(f) anchor preservation (c6/c22/c26/c36/c38 read-only)	NOT YET APPLICABLE

Validators (promise_check, org_check) not run this range — ledger state is as-of _run/cycle_44_ closed (ledger 727, 0 ERRORS per c44 close), no new emissions since; running them now would surface last-cycle state, not this cycle’s deliverable. Will run next cycle against the actual completed emission set.

MINOR Process Observations (Logged)

1. **Cycle-49 Hold Pattern researcher brief:** authored "Standing by for the c45 fanout directive... no in-flight work items on this clone." This is the exact Hold Pattern anti-pattern the operating protocol names — authoring a cycle whose sole deliverable is a pause memo conditioned on external ratification. Directive authority is already granted by the campaign prompt; no user to wait on. Worker corrected proactively at cycle 50 by advancing on the c26 Path B pre-registered plan when audio arrived, which is the right response to a Hold-Patterned brief.

2. **Second observed instance recently** of a researcher emitting a “standing by” brief with no substantive directive. If this recurs, propose plan-level tightening: `research_brief` without a `<parallel_cycle_fanout>` block or a linear scope statement is a plan-of-record violation, not a valid cycle input. Not urgent; log for pattern-tracking.

Egress State (Documented; Provenance Note for Report)

Worker’s egress probe recorded `429 + tv_embedded closure`. This is NOT the two-consecutive-`media_ok=true unblock` signal. The 49 clips being extracted must therefore be from already-on-disk audio (partial prior harvest, or manually-seeded corpus growth since c38’s 43-song frozen snapshot). Audio arrivals since c38 must be from a channel outside the automated harvest — worth documenting the provenance chain in the eventual report.

This suggests the on-disk rated corpus has grown modestly since c38’s 43-song frozen snapshot. Real Path B progression, not null retry, **provided pre-registration discipline holds**.

Fanout Cadence Guidance (Cycle-51 Auditor)

Do NOT open a c45 fanout while c45’s ear_v2 work is still in flight. Complete the single substantive milestone first; a fanout across three independent branches when one is mid-extraction violates the `parallel_cycle_fanout` independence self-check.

If the current cycle resumes as this same clone: finish `ear_v2` with rubric-first ordering. If a fresh cycle-45 opens: the primary directive is “complete M-EAR-1/real-label-training-v2 per items 1-5 above”. The researcher’s role is to write that brief, not to stand by.

Ledger Events (This Range)

Zero substantive events. No `M-EAR-1/real-label-training-v2/*` event emitted (milestone unregistered in plan-of-record; rubric unwritten; verdict absent). Housekeeping deferred to the completion cycle. Ledger state as-of `_run/cycle_44_closed` at 727 events with 0 `promise_check` ERRORS; no new emissions since.

State-Machine Discipline (c29 Lemma Respected)

`M-EAR-1/real-label-training-v2` is a peer sub-milestone under `M-EAR-1` (would fire in-progress on first substantive event at completion cycle). NOT a child of terminal-validated `_manager/M-EAR-1-path-B-commit`, `M-EAR-1/{synthetic-label, head-regularization, feature-representation}-audit`, `M-EAR-1/armed-harness-fixture-reinforcement`, c36 v0, or c38 v1. The c42 closure event `_manager/M-EAR-1-real-label-training-v2-unfixable-by-audit-clone-1` (opened in-progress; first emission on new identifier per c29 lemma) may transition to validated at the sequential-mode completion or remain durable as a historical marker; c45 completion cycle owns the disposition.

Zero `[[BRANCH_COMPLETE]]` emitted; milestone entirely in-flight from a sequential-mode perspective.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.
- SHA-256 tiebreak; no PRNG; no `sidecar_nonfactor` imports; no `i4_stratified` import.
- Interpreter guard `/usr/bin/python3` on any new script; c42-preserved scripts/`ear_v2/*.py` AST-clean vs c22/c23/c25 forbidden imports (verified through c41).
- Read-only anchors preserved (nothing touched this range beyond feature-cache growth):

c6 feature cache + CORN chassis; c22 stability harness; c26 Path B commitment; c36 v0; c38 v1; c1 chunker; c42-preserved 203/252 .npy cache serving as the sequential-mode extractor's cache-skip base.

- c26-frozen SB1/SB2/SB3 thresholds unchanged; no post-hoc adjustment attempted.
- Rated audio egress-blocked at *.googlevideo.com (429 + tv_embedded closure this cycle; retry cadence at conductor level; 49 newly-cached clips from operator-seeded on-disk arrivals since c38, provenance chain to be documented in eventual report).
- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses
UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).

Anti-Patterns Locked (5-Count Stable; Path B v2 Not Under Path A Locks)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. c31 STILL_GAP / c35 A anti-pattern surface intact. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged.

Note: last three anti-patterns are all Path A ear-model exhaustion — v2 is Path B and does not fall under those anti-patterns. Corpus-side intervention (anchored-tail per-song resample) remains orthogonal to c22/c23/c25 axes.

Hold Pattern anti-pattern surfaced (MINOR) at cycle 49 researcher brief; corrected by worker's cycle-50 proactive advance. Not blocking; logged for pattern-tracking.

Null-cycle anti-pattern correctly avoided at cycle 51: no VALIDATED emitted on a cycle producing zero substantive deliverable while real work runs.

Root-Conductor Sequential-Mode Handoff (In-Flight; Per c42 Closure)

Sequential-mode pickup consuming the c42-preserved handoff assets is now empirically in flight (contra c41 PIVOT's expectation that this work would move outside fanout scope):

1. **Feature extraction:** 49 additional clips extracting on top of c42-preserved 203/252 .npy cache. Newly-cached clips are from operator-seeded on-disk arrivals since c38's frozen snapshot (429 egress probe confirms harvest still blocked).
2. **scripts/ear_v2/train_v2.py:** awaiting extraction completion; 5-fold GroupKFold; per-song grouping preserves no-clip-leakage contract.
3. **scripts/ear_v2/evaluate_sb_v2.py:** awaiting rubric-first commit + train completion; c26-frozen thresholds unchanged.
4. **scripts/ear_v2/determinism_check.py × 2:** standard tail-of-pipeline verification on training_result.json + corn_head_v2.pt.
5. **tests/test_ear_real_label_training_v2.py:** ≥14 tests scaffolded per c41; awaiting first invocation post-extraction.
6. **6 substantive + 4 housekeeping ledger events** originally scoped under M-EAR-1/real-label-training-v2/* (no -clone-* suffix in sequential mode — c33 harness guard applies only in clone context per c39 v3 doc codification).

Cycle-52 Handoff (Priority Order, Per Cycle-51 Auditor)

Primary (single-item):

1. **Complete M-EAR-1/real-label-training-v2 per items 1-5 of MODERATE flag above.** Rubric-first ordering; register plan row; commit rubric doc + SHA; run train_v2 → evaluate_sb_v2 → determinism_check × 2 → tests → report → ledger events. Do NOT open a c52

fanout until ear_v2 completes.

Deferred (opportunistic):

- `_infra/worker-cli-startup-health-check-clone-*` (from c43 range) — pre-spawn readiness probe.
- `M-RULES-1/extraction/rated-corpus/harmonic-window-refinement` (from c40 clone-0) — standing.
- **Band-6** `f1cfe4855364ea9b` focused-rerun from c39 auditor — standing.
- `_infra/emitter-idempotence-guard-clone-*` — standing.
- `_manager/effects-chain-band-selectivity` — opportunistic.
- **c38 clone-1 REDEFINED_GAP + normalizer-v2 REFUTED** mscore3 quantization root-cause narrowing — opportunistic.
- **c37 VST3 activation** still gated by c36 MIXED verdict.
- `_manager/fanout-pipeline-cost-audit` — enumerate which M-* milestones exceed fanout-cycle capacity.
- **Plan-level tightening ticket:** `research_brief` without `<parallel_cycle_fanout>` block or linear scope statement should be a plan-of-record violation. Not urgent; recurrence-triggered.
- **Egress retry** per campaign directive.

Cumulative Progress

M-EAR-1 arc (post-cycle-49-51 sequential-mode pickup entry):

Cycle	Milestone	Status
c22-c25	Path A chassis chain	insufficient (anti-patterns locked)
c26	<code>_manager/M-EAR-1-path-B-commit</code>	committed; three SBs frozen
c31	<code>armed-harness-fixture-reinforcement</code>	FIXTURE_READY
c36	<code>real-label-training-v0</code>	EAR_v0_INSUFFICIENT (43/80)
c37 clone-1	<code>_manager/ear-sb3-statistic-degeneracy-fallback-statistic</code>	F1_ADOPTED
c38 clone-0	<code>real-label-training-v1</code>	EAR_v1_PARTIAL (43/80)
c39-c41 clone-1	<code>real-label-training-v2</code> (fanout attempts 1-4)	PARTIAL_PROGRESS → PIVOT (unfixable-by-audit)
c42 clone-1	<code>closure sequence for real-label-training-v2</code>	COMPLETE with <code>[[BRANCH_COMPLETE]]</code> ; 203/252 cache preserved for root-conductor sequential-mode pickup
c45 sequential-mode pickup (this range)	<code>real-label-training-v2 sequential-mode extraction</code>	IN-FLIGHT → CONTINUE (49 additional clips advancing; rubric-first pre-registration correction owed before <code>evaluate_sb_v2</code>)

Real Path B progression, not null retry: on-disk rated corpus grew modestly since c38's 43-song frozen snapshot; operator-seeded arrivals; 49 clips additive to c42-preserved 203/252

cache. Egress harvest still blocked (429 + tv_embedded closure this cycle); the two-consecutive-media_ok=true unblock signal has not fired.

Sequential-mode pickup empirically working (contra c41 PIVOT's expectation that this work would move outside fanout scope; the c42-preserved handoff assets are being consumed as designed).

Discipline arc c37→c44 held: rubric-hash three-way byte-equality; mtime-strict-order gate; ≥ 30 SHA anchor preservation; byte-determinism $\times 2$ in fresh mkdtemp; peer-shard-only-on-LANDS; anti-cheat identity-cell test; ≥ 15 -case test suite with anchor-invariant tests. **Cycle 52 must preserve this streak** by committing rubric SHA before evaluate_sb_v2 runs; MOD-ERATE flag proactively raised at cycle 51.

c29 state-machine lemma respected: no validated $\rightarrow$ in_progress transitions; no [[BRANCH_COMPLETE]] emitted on in-flight milestone; c42 _manager/*-unfixable-by-audit-clone-1 marker durable as historical artefact.

c32 $\rightarrow$ c33 $\rightarrow$ c36 v2 $\rightarrow$ c39 v3 fanout-namespace convention held vacuously (zero events emitted this range; sequential-mode has no -clone-* suffix per c33 harness guard being clone-context-only per c39 v3 doc codification).

Collision-modeling arc: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Ledger state: 727 events (as-of _run/cycle_44_closed); 0 promise_check ERRORS; no new emissions since. c45 completion cycle will resume the emission stream with the rubric-committed event plus subsequent per-stage events per the codified post-artefact discipline (c39 Issue #6).

Cycle economics: this range consumed audit turns but produced zero substantive deliverable while real work (feature extraction) advances in the background. Direction sound; completion cycle should prioritise rubric-first ordering correction + finish the pipeline in a single tool-call chain per c42-proven discipline.

[END OUTPUT]